Yonwoo Choi

Algorithm Engineer
GenGenAI

Email: yonwoo@gengen.ai, **Web**: yc4ny.github.io, **Google Scholar**: Link

EDUCATION

Seoul National University *2024*
M.S. in Computer Science and Engineering
Advisor: Prof. Hanbyul Joo
Thesis: A Wrist Mounted Camera System for Hand Pose Estimation from Single RGB Images

University of Virginia *2021*
B.A. in Computer Science

RESEARCH INTEREST

My research lies in 3D computer vision centered on humans, with a focus on capturing, understanding, and reconstructing people in 3D. Ultimately, I aim to build machines that perceive and understand human motion and behavior as naturally as people understand one another. I pursue this direction by building practical capture systems and learning-based methods that bring humans into the digital world.

PUBLICATIONS

*: equal contribution, †: corresponding author

"LEGO: A Lifting-Free Approach for Exocentric-to-Egocentric Video Generation"
Suhwan Cho*, **Yonwoo Choi***, Soongjin Kim*, Jicheol Park, Taegyu Lim†
arXiv preprint arXiv:2610.12442, 2026.

"SVAD: From Single Image to 3D Avatar via Synthetic Data Generation with Video Diffusion and Data Augmentation"
Yonwoo Choi
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**) Workshop, 2025.

WORK EXPERIENCE

GenGenAI, South Korea
Algorithm Engineer (Alt. Military Service) *Sept. 2025 – Present*
Worked on Controllable Image and Video Generation Across Domains

SECERN AI, South Korea
Research Engineer (Alt. Military Service) *Mar. 2024 – Sept. 2025*
Worked on Animatable 3D Human Avatars

Visual Computing Lab, Seoul National University
Graduate Research Assistant *Mar. 2022 – Feb. 2024*
Advisor: Hanbyul Joo
Worked on 3D Egocentric Hand and Body Pose Estimation

Undergraduate Research Opportunities Program, University of Virginia
Undergraduate Researcher (for credit) *Oct. 2020 – May 2021*
Advisor: Raymond Pettit
Worked on Automated Assessment Tools

OPEN SOURCE

LEGO: `https://github.com/suhwan-cho/LEGO`
Official implementation of the paper LEGO

SVAD: https://github.com/yc4ny/SVAD
Official implementation of the paper SVAD

Facebuilder: https://github.com/yc4ny/facebuilder
Multiview face modeling with real-time 3D consistency

SMPL Fitting: https://github.com/yc4ny/SMPL-Fitting
Parametric human shape and pose optimization

Egocentric Human Pose Estimator: https://github.com/yc4ny/xR-EgoPose
Egocentric 3D human pose estimation from a single fisheye camera

Multiview Human Pose Capture System
https://github.com/yc4ny/Multi-Camera-3D-Human-Reconstruction
5 camera system setup for 3D human reconstruction

PROJECTS

Missile-Perspective Naval Scene Video Generation *Dec. 2025 – Dec. 2026*
Missile-perspective synthetic video generation for naval target detection
GenGenAI, Hanwha Aerospace

Missile-Perspective Ground Scene Video Generation *Mar. 2026 – July. 2026*
Missile-perspective synthetic video generation for ground vehicle detection
GenGenAI, Hanwha Aerospace

Multiview 3D X-Ray Detection for Airport Screening *Mar. 2024 – Sept. 2025*
3D X-ray calibration, 3D bounding box detection
SECERN AI, Ministry of Land, Infrastructure and Transport

TEACHING

Teaching Assistant, Seoul National University *Fall 2023*
4190.408 Artificial Intelligence (Instructor: Hanbyul Joo)

Teaching Assistant, Seoul National University *Spring 2022*
4190.773 3D Computer Vision (Instructor: Hanbyul Joo)